



University of AJ&K Neelum Campus

Department of Computer Science and IT

Computer Organization and Assembly Language
Lab Book

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1 Hello World in Assembly

1.1 Description

This is a simple assembly language program that displays the message "Hello, from Assembly!" on the screen.

1.2 Code

Hello World Program

```
.686
INCLUDE Irvine32.inc
.data
    msg BYTE "Hello, from Assembly!",0
.code
    Main PROC
        lea edx, msg
        Call Clrscr
        Call WriteString
        EXIT
    Main ENDP
END Main
```

1.3 Output

Output

```
Hello, from Assembly!
```

2 Adding Two Numbers

Hello World Program

```
.686
INCLUDE Irvine32.inc

.data
    number1 BYTE 10
    number2 BYTE 10
.code
    Main PROC
        mov al, number1
        add al, number2

        movzx eax, al
        Call Clrscr
        Call WriteInt
        EXIT
    Main ENDP
END Main
```